

Scratch

Firm type (`is_biotech`, `is_pharma`)

I initially planned to construct binary firm-type indicators using external biotech/pharma classifications. However, these categories are not mutually exclusive or consistently defined: many firms operate across both biotechnology and traditional pharmaceutical development. I therefore avoided imposing a potentially subjective firm-type label and used annual revenue as a continuous, consistently observable measure of firm size.

SIC codes as a biotech classifier

I initially considered using SIC codes to distinguish biotechnology from pharmaceutical firms. However, the SIC system was developed in the 20th century and its industry taxonomy predates the emergence of modern biotechnology as a distinct sector. As a result, codes such as SIC 2834 (Pharmaceutical Preparations) pool heterogeneous firms ranging from small R&D-focused biotechs to multinational pharmaceutical incumbents. Conversely, the unfiltered sponsor universe contained non-pharmaceutical firms such as Coca-Cola. I therefore used SIC codes as a sample-selection/industry filter, rather than a firm-type classifier, and used annual revenue as the primary measure of firm size.

Hand-built biotech/pharma stock lists

I initially considered constructing `is_biotech` and `is_pharma` by manually identifying publicly traded biotech and pharmaceutical companies from stock-market/company lists. This would have provided a more intuitive firm-type classification than SIC codes, but it would require subjective decisions about which companies qualify as biotech versus pharmaceutical and would be difficult to apply consistently across the full 2016–2026 period.

Private and non-U.S.-listed firms

Excluded due to data availability. I considered expanding the sample to private and non-U.S.-listed firms using alternative sources such as Wikidata and NLP-based inference from sponsor descriptions. I ultimately rejected these approaches because I wanted firm size to be based on a single, standardized, publicly auditable financial source rather than inferred values with uncertain measurement error. This creates an important coverage limitation: private and non-U.S.-listed firms are likely to be disproportionately represented among smaller and earlier-stage companies, meaning their exclusion may be systematically related to both the firm-size classification and the clinical-development outcome.

Revenue deflation and R&D-spending criterion

I considered both adjusting revenue for inflation and incorporating R&D spending into the definition of small biotechnology firms. Both were feasible extensions, but were excluded to maintain scope and because of time constraints. These are therefore among the first methodological extensions I would prioritize in future analysis.

Global revenue rankings / top-20 pharmaceutical companies

Rejected. I considered defining large pharmaceutical firms using a global revenue ranking or a hand-assembled list of the largest pharmaceutical companies. I instead used a published revenue threshold so that firm classification would be based on a transparent and reproducible rule, and to preserve meaningful variation in firm size within the analysis.

Employee headcount

I rejected this as a measure of firm size as, although employee headcount may conceptually capture organizational scale, it can change substantially. More importantly, consistently measured point-in-time headcount was not available across the full set of firms and years required for the time-varying design. Annual revenue therefore provided a more consistently available measure for this analysis.

Risk defined by drug novelty

I considered defining development risk using drug novelty or first-in-class status rather than clinical-trial phase. This would provide a more direct measure of development risk, but the available registry data were insufficient to construct a reliable novelty measure. I therefore use clinical-trial phase as a measure of development stage, rather than treating phase as a direct measure of risk.

Therapeutic indication

Therapeutic indication is potentially relevant because the mix of indications could influence the relationship between firm size and clinical-trial phase. However, I did not have sufficient time nor enough knowledge to analyze and normalize the AACT conditions data within this study. I therefore excluded indication from the current analysis and treat it as an important area for future work.

Formal testing

Due to time constraints, I did not conduct formal hypothesis testing, regression analysis, or other causal/statistical approaches to test the relationship between firm size and trial phase. Instead, I focused on a descriptive comparison of phase distributions.